

MEMORANDUM - CITY OF PASADENA
DEPARTMENT OF PUBLIC WORKS

DATE: August 18, 2021

TO: Public Safety Committee

FROM: Kris Markarian, P.E., Acting Director of Public Works

SUBJECT: Colorado Street Bridge Barrier Enhancements Project –
Recommendation of Barrier Design Option

RECOMMENDATION:

It is recommended that the Public Safety Committee receive this update report on the Colorado Street Bridge barrier enhancements and consider how best to proceed.

BACKGROUND:

On February 3, 2020, staff presented to the Public Safety Committee the Colorado Street Bridge vertical barrier design concepts which received the most public support for the Committee's review and input.

Vertical barrier concepts for the Colorado Street Bridge were developed by the City's consultant, Donald MacDonald Architects and reviewed for general conformance to the Secretary of Interior Standards. These concepts were developed through a public outreach process that included input from Pasadena Heritage and the project's Task Force members over the course of several meetings held between August and November of 2019. On November 19 and 26, 2019, these concepts were presented to the Historic Preservation Commission and Design Commission, respectively, for advisory review and comments.

In line with both commissions' feedback, the Public Safety Committee requested that staff construct full-scale mock-ups of the preferred designs for better visualization and evaluation by the public before a specific design concept could be chosen.

Councilmember Kennedy also asked staff to explore concepts for full enclosure of the bridge deck.

In June 2017, a temporary 10-foot tall chain-link fence was installed at each of the bridge alcoves. On September 2, 2018, after a 13-hour incident when first responders successfully persuaded an individual to come down from the bridge ledge, City Manager issued an emergency purchase order for the installation of a new 10-foot chain-link fence spanning the entire length of the bridge.

Tables below represent data from Pasadena Police Department, tabulated from January 1, 2015 to June 13, 2021:

YEAR	Climbed/Attempted to Climb	Didn't Climb (standing near fence; contemplating)	Breached Fence (over the balustrade)	Between Fence (behind temp. chain-link fence in alcove)	Total Incidents
2015	3	3	6	N/A	12
2016	2	10	17	N/A	29
2017	5	14	13	N/A	32
2018	4	12	7	1	24
2019	8	14	3	1	26
2020	2	8	1	4	15
2021	0	2	1	2	5

YEAR	SUICIDES
2015	4
2016	2
2017	10
2018	4
2019	1
2020	0
2021	1

PROJECT UPDATE:

Pursuant to the request of the Public Safety Committee on February 3, 2020, Public Works hired a contractor and completed the fabrication and installation of full-scale mock-up panels on the Colorado Street Bridge.

On March 3, 2021, the following three (3) concepts were erected at two locations along the bridge for public viewing and assessment: (See Attachment A)

- Option A – Mesh (Curved-Straight)
- Option B – Mesh (Curved-Curved)
- Option C – Picket

Upon installation of the mock-up panels, Public Works took the following steps to present the potential design solutions and solicit community opinion and feedback:

- (1) Community Outreach – A Virtual Open House meeting was held on March 25, 2021 to present the full-scale mock-up panels installed on the Colorado Street Bridge and introduce the web survey to the community.
- (2) Pasadena Heritage – On April 15, 2021, Public Works staff met with Pasadena Heritage staff and volunteers on the Colorado Street Bridge to closely review the

mock-ups and discuss possible refinements or modifications of each option and their recommendations for the vertical barrier.

- (3) Commissions - On April 20, 2021 and May 11, 2021, staff and Donald MacDonald Architects presented the mock-ups to the Historic Preservation Commission and Design Commission, respectively, for advisory review and comments.

The majority of Historic Preservation Commission members voted to recommend Option B – Mesh (Curved-Curved) due to its effectiveness and preservation of the bridge with the following considerations:

- Minimize framing dimensions or proportions where possible;
- Incorporate more architectural features characteristic to the bridge, such as arches, into the panels; and
- Ensure the globe fixtures on the light poles are above the barrier and are unobstructed from view.

The consensus among the Design Commission was that staff should further study concepts that incorporated more transparency at the alcoves and light poles and maintained the light pedestal height. The commissioners preferred the straight mesh panels (similar to Option A) between the alcoves, with an added gap between the bottom of the panels and the top of the balustrade.

- (4) Online Survey – A public survey was published on the project's website from March 4, 2021 to May 31, 2021 to solicit public opinion on the mock-up design options. The community was notified of the survey through press releases and City social media postings. Postcards were also distributed to 583 residences within a 1,000-foot radius from the bridge.

The City received 2,268 responses to its online survey. The following table is a summary of survey data indicating the most preferred mock-up design option:

Number of Votes Ranking Options	1,038*
Option A – Mesh (Curved-Straight)	216 (20.8%)
Option B – Mesh (Curved-Curved)	462 (44.5%)
Option C – Picket**	360 (34.7%)

**Received 2,268 responses to survey, but only 1,038 ranked the mock-ups.*

***59% of respondents listed Option C as their least favorite.*

622 comments were received of which 44% preferred barriers not be installed and the bridge returned to its original state (without temporary chain-link fence).

Demographics of the voters:

- 2,112 have not attended to any public meetings/presentations regarding the Colorado Street Bridge
- 1,766 live near the bridge
- 1,246 view the bridge when driving by
- 1,385 travel over the bridge
- 647 view the bridge from below (park or trail)

Based on the comments received from Historic Preservation Commission and Design Commission, supplementary renderings were developed to illustrate variations to the concepts that emphasize transparency around the light poles and alcoves. (See Attachment B)

In response to Councilmember Kennedy's request, alternative concepts were explored for fully enclosing the bridge. Renderings have been developed to illustrate this preliminary concept, but have not been vetted through public outreach or presented to Historic Preservation and Design commissions. This preliminary concept was presented to the project's Task Force members, including Pasadena Heritage and first responders from City's Police and Fire departments, who did not deem it to be acceptable from an aesthetic or safety perspective. (See Attachment C)

Alternative vertical barrier options to those presented in Attachment A were also considered by the project's task force, but were not deemed viable due to various reasons including effectiveness, aesthetics, and required maintenance.

CONSIDERATIONS FOR NEXT STEP:

To date, the preliminary design and a draft of the California Environmental Quality Act (CEQA) initial study and mitigated negative declaration (IS/MND) have been completed. Upon City Council's approval of a barrier concept design, staff will complete the project design, and prepare IS/MND report and construction documents.

From a functional point of view, Option A and Option B provide approximately the same level of protection. The main difference between the two options is largely aesthetic. To date, a total of \$1,050,000 has been expended on the design of suicide barriers for the Colorado Street Bridge. There are likely other potential barrier options which could be explored, however, additional budget appropriations would be needed. Moreover, the ultimate cost to install either Option A or B is likely to be in the \$4-6 million range, and depending on the complexity of any other selected option, the cost could be higher.

Given all these factors, the Public Safety Committee is asked to consider whether to recommend to the City Council, proceeding with the implementation of either Option A or B, leaving the currently installed temporary fencing in place for the time being, or taking some different action.

FISCAL IMPACT:

To date, City Council has appropriated \$1,050,000 to the *Colorado Street Bridge Barrier Enhancements (73324)* CIP project which has been expended for the following:

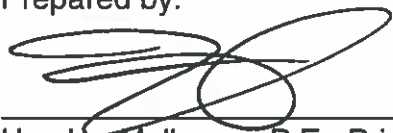
- Temporary 10-foot tall chain-link fence installation and maintenance
- Mock-up panel design and installation
- Preliminary Environmental Assessment and Conceptual Design, including public outreach

Based on where the project is in the preliminary design phase, Staff estimates the cost for completion of design and construction at \$4 - \$6 million. Once Council determines the final design option, a more definitive and detailed cost estimate and fiscal impact analysis will be performed and presented to City Council.

Respectfully submitted:


Kris Markarian, P.E., Acting Director
Department of Public Works

Prepared by:


Hayden Melbourn, P.E., Principal Engineer
Department of Public Works

Attachments:

- A – Vertical Barrier Design Options A - C (*presented to Public Safety Committee on February 3, 2020*)
- B – Refined Options (*pursuant to Historic Preservation Commission and Design Commission comments received April 20, 2021 and May 11, 2021, respectively*)
- C – Alternative Concept – Cable Net Enclosure of Bridge Deck
- D – Comparative Tables of Options

**Attachment A –
Vertical Barrier Design Options A - C
(*presented to Public Safety Committee
on February 3, 2020*)**

OPTION A – MESH (CURVED –STRAIGHT)

[Curved barrier at every lamp post and alcove, straight barrier along remaining length of bridge].



View from the bridge



View of the bridge



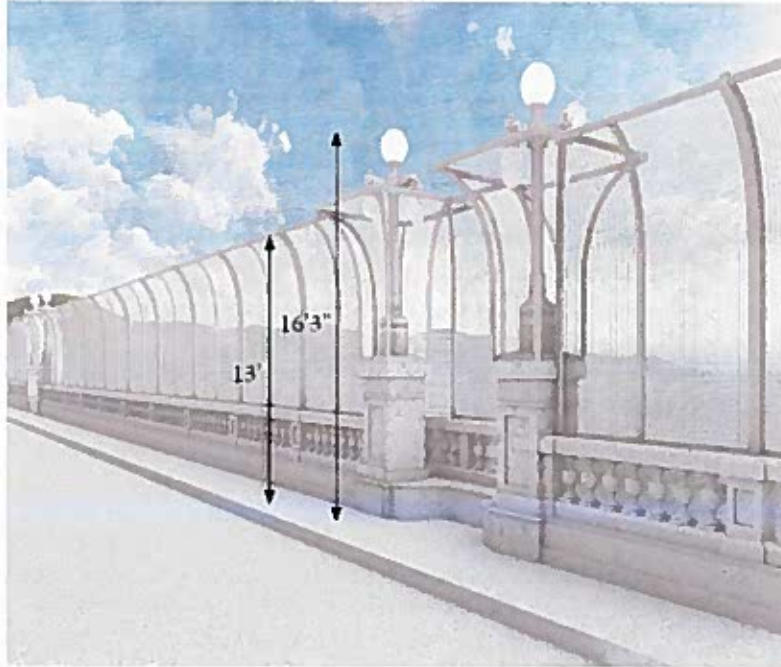
NOTE:

- *The red support is temporary and not a part of the design*



OPTION B – MESH (CURVED-CURVED)

[Curved barrier over entire length of bridge, including every lamp post and alcove]



View from the bridge

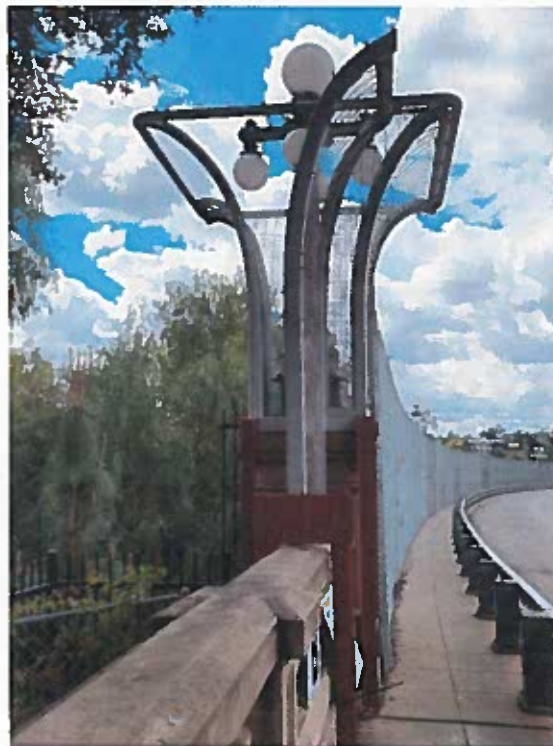


View of the bridge



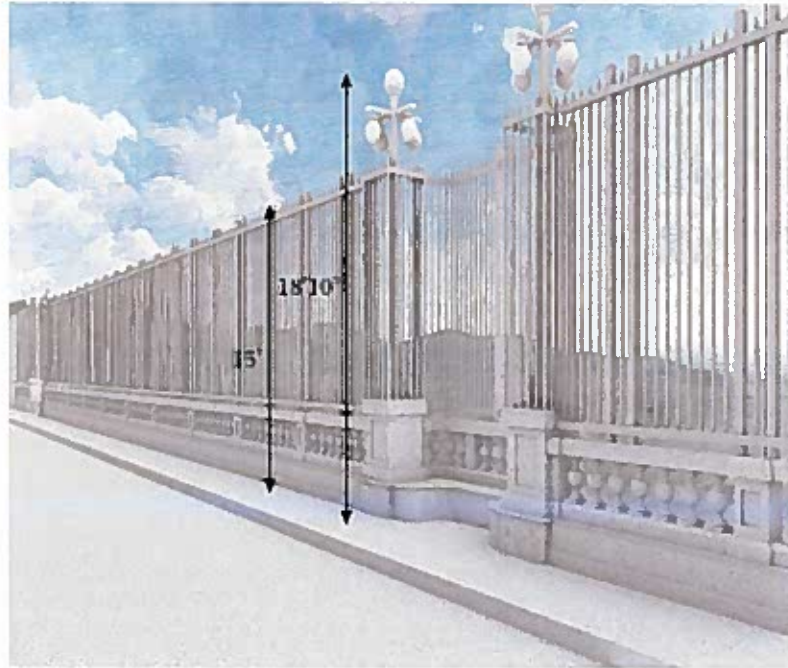
NOTE:

- *The red support is temporary and not a part of the design*



OPTION C – PICKET

[Vertical/straight barrier over entire length of bridge, including every lamp post and alcove]



View from the bridge



View of the bridge



NOTE:

- *The red support is temporary and not a part of the design*



BARRIER MATERIALS

The following images show different materials proposed for the barrier. Materials under consideration include stainless steel mesh fabrics and metal pickets. Mesh fabrics provide the best transparency and are very effective at deterring climbing by reducing handholds and footholds. Pickets require a taller barrier to achieve a 10-foot distance between the horizontal top and bottom rails and are more easily climbed. These materials and their support frame configurations are shown below:



DOGLA TRIO



DOKA BARRETTE

Attachment B–
Option Variations/Refinements
(pursuant to Historic Preservation Commission
and Design Commission comments
received April 20, 2021 and May 11, 2021, respectively)

OPTION A (Panel Variation) – MESH (CURVED –STRAIGHT)

[Curved barrier at every lamp post and alcove, straight barrier along remaining length of bridge].



OPTION A (Hybrid)– MESH (CURVED –STRAIGHT)

[Curved barrier at every lamp post and alcove, straight barrier along remaining length of bridge].



OPTION A Straight Panel– MESH (STRAIGHT-STRAIGHT)

[Vertical/Straight barrier over entire length of bridge]

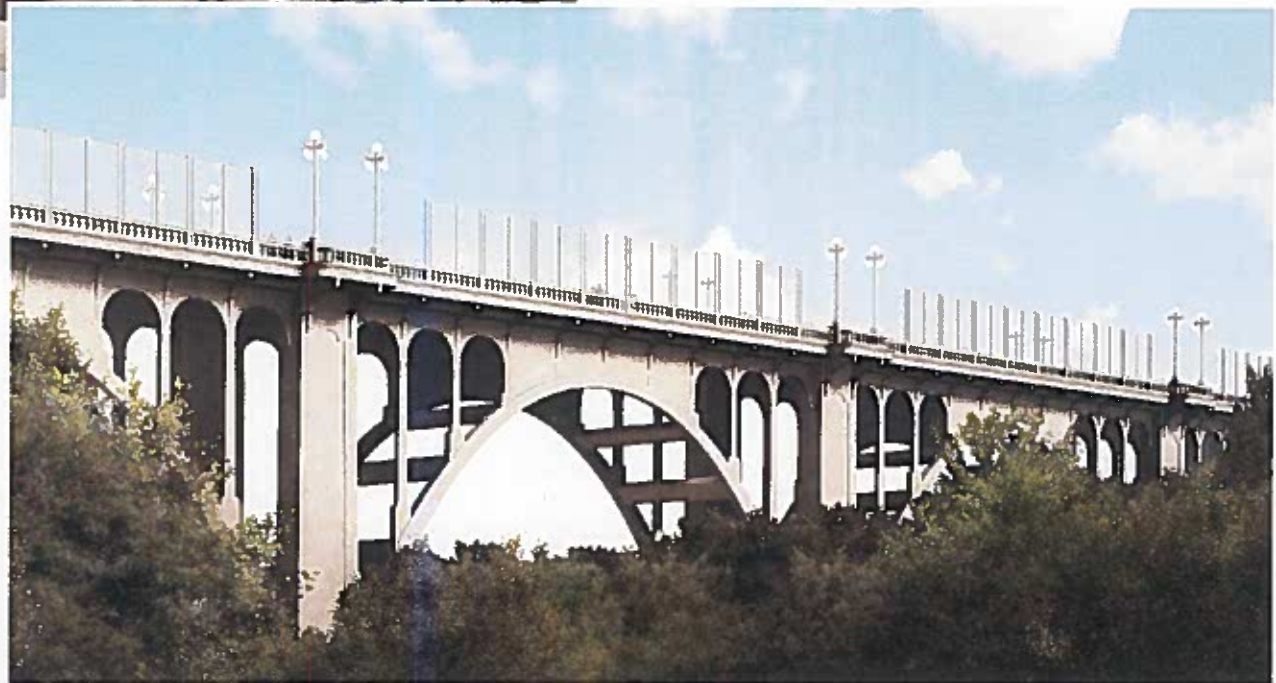


EXPLORATION OF TRANSPARENT PANELS AT ALCOVES

REFINED OPTION 01 – GLAZED ALCOVE AND MESH PANELS



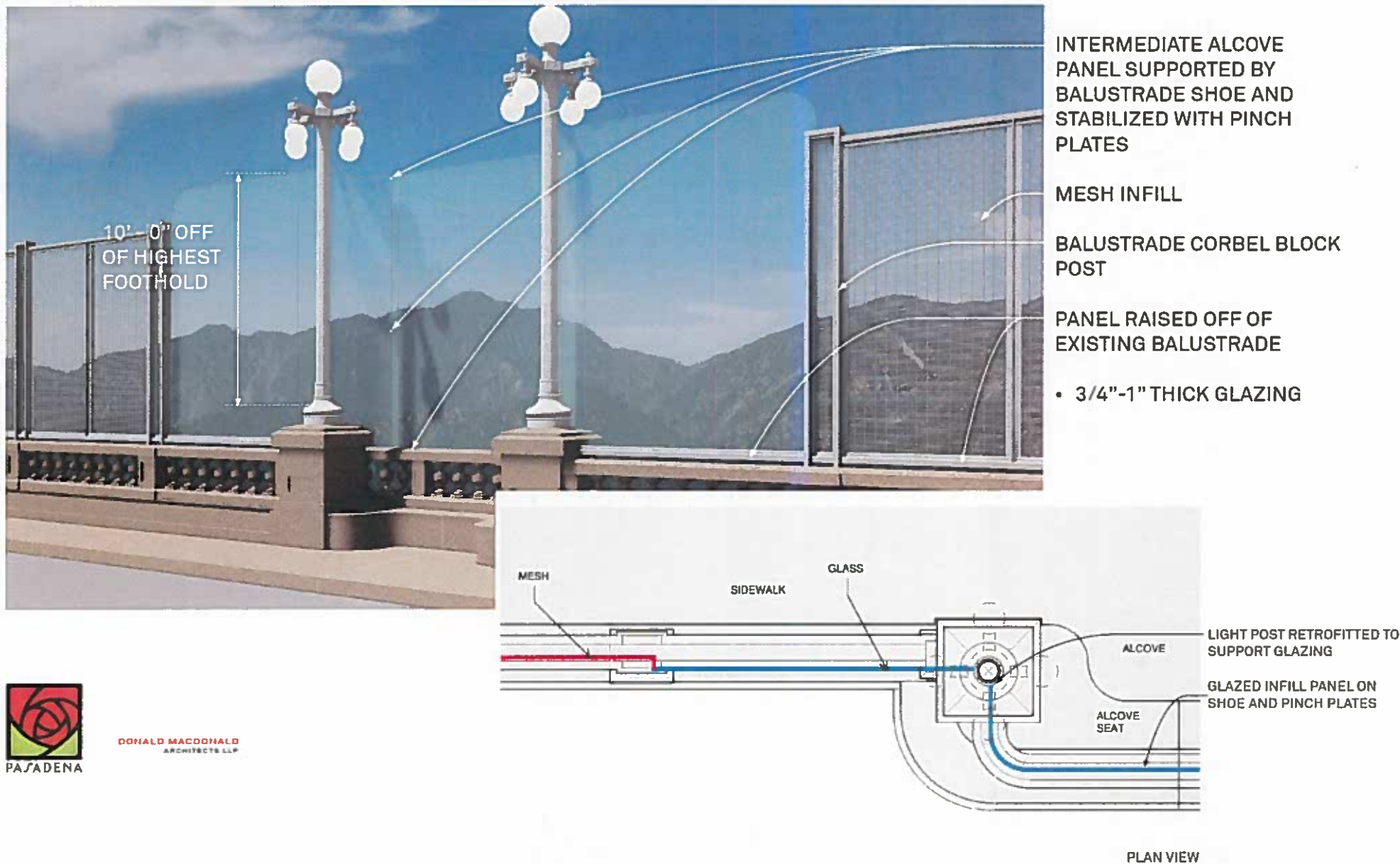
SHAWN D. MACDONALD
ARCHITECTS LLP



PARK PERSPECTIVE

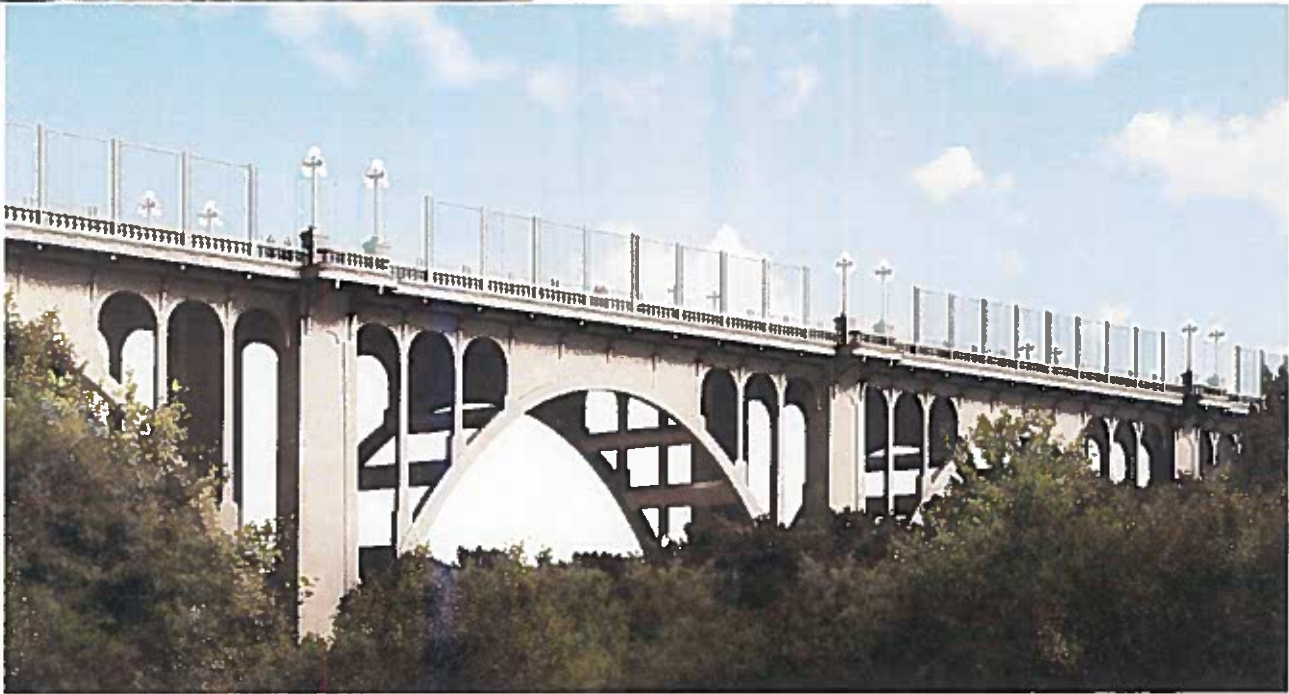
EXPLORATION OF TRANSPARENT PANELS AT ALCOVES

REFINED OPTION 01 – GLAZED ALCOVE AND MESH PANELS



EXPLORATION OF TRANSPARENT PANELS AT ALCOVES

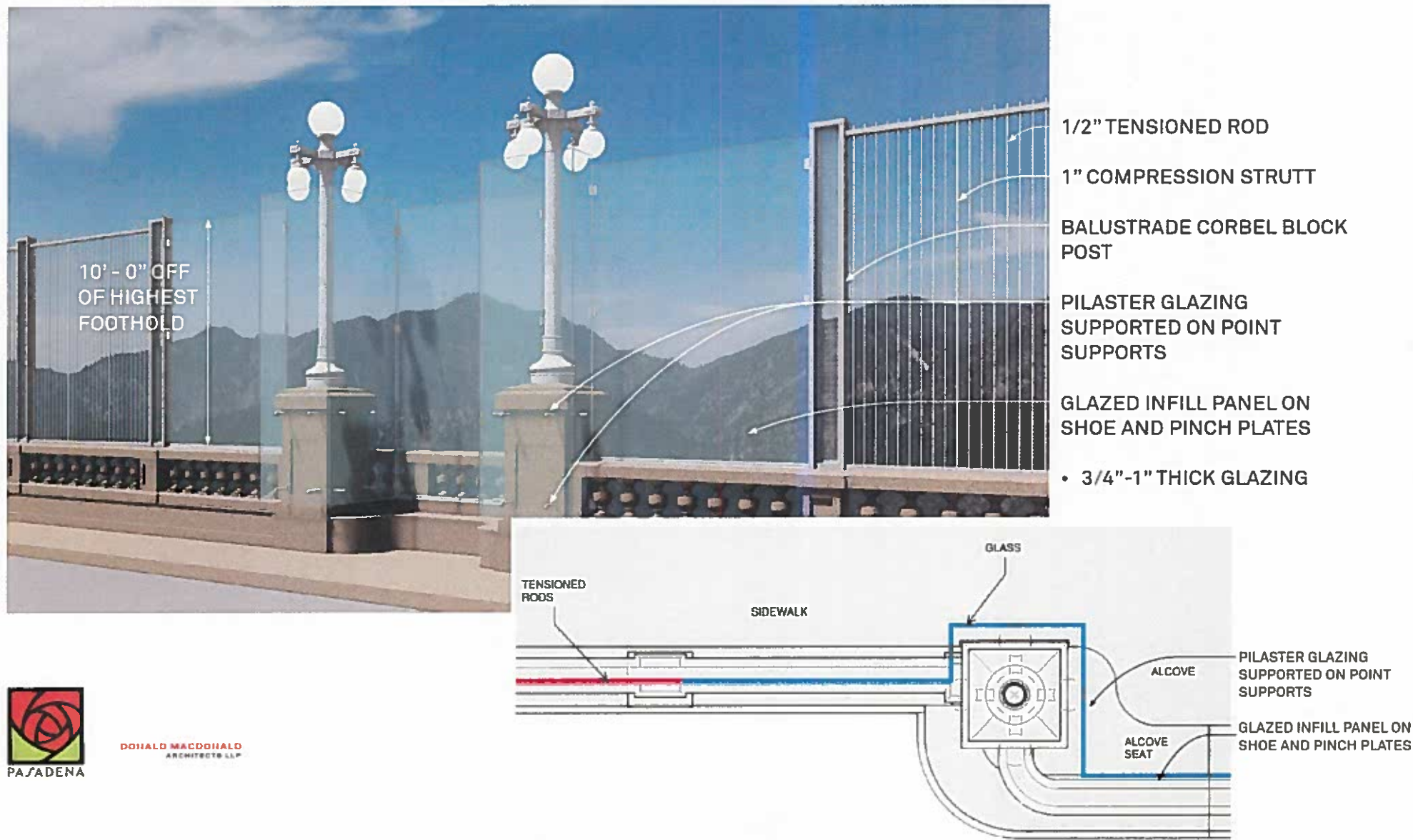
REFINED OPTION 02 – GLAZED ALCOVE AND TENSIONED RODS



PARK PERSPECTIVE

EXPLORATION OF TRANSPARENT PANELS AT ALCOVES

REFINED OPTION 02 – GLAZED ALCOVE AND TENSIONED RODS

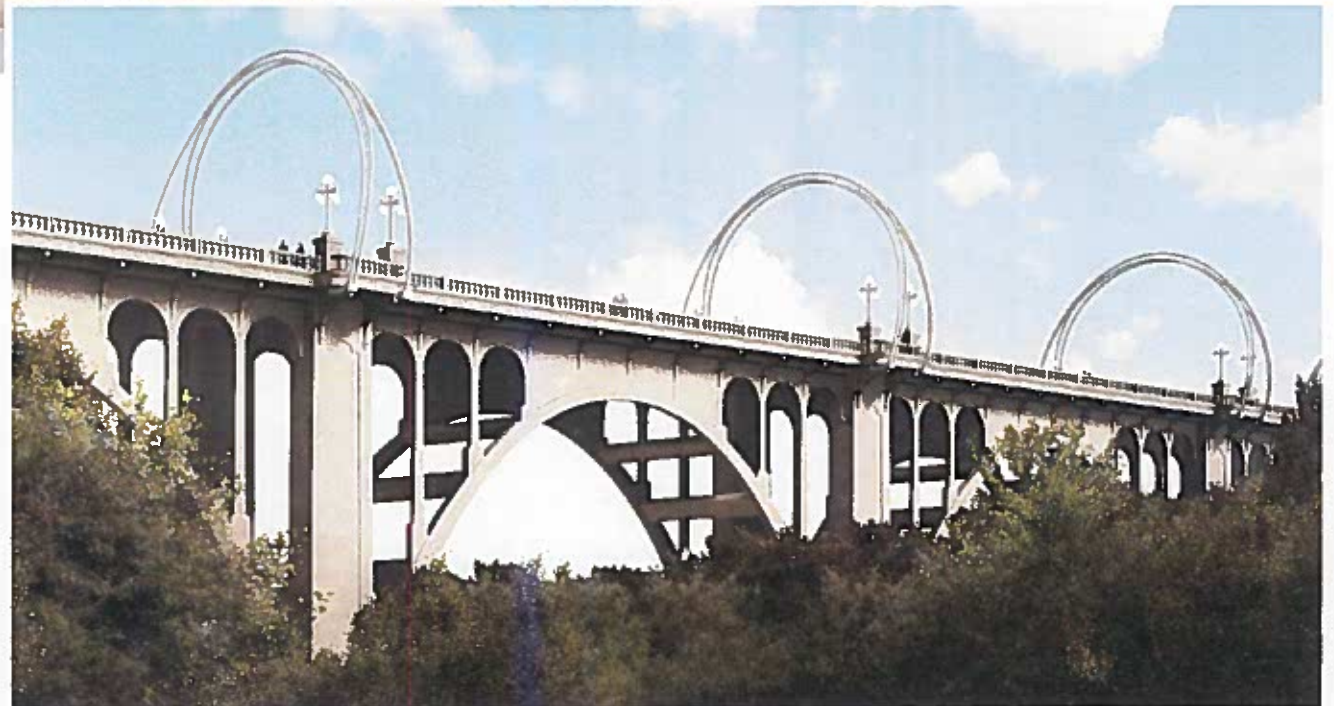


**Attachment C–
Alternative Concept
Cable Net Enclosure of Bridge Deck**

CABLE NET ENCLOSURE SUPPORTED ON STEEL HALOS



OFFICE OF MUNICIPAL
AFFAIRS



PARK PERSPECTIVE

CABLE NET ENCLOSURE SUPPORTED ON STEEL HALOS



MULTI-PIPE STEEL 'HALO'
BIFURCATED AT ALCOVE TO
FRAME VIEWS OUT

CABLE NET ENCLOSURE

SUSPENSION CABLE

EXISTING FENCE TO BE
REMOVED

- LAMPS AND PILASTERS
REMAIN AT EXISTING
HEIGHT

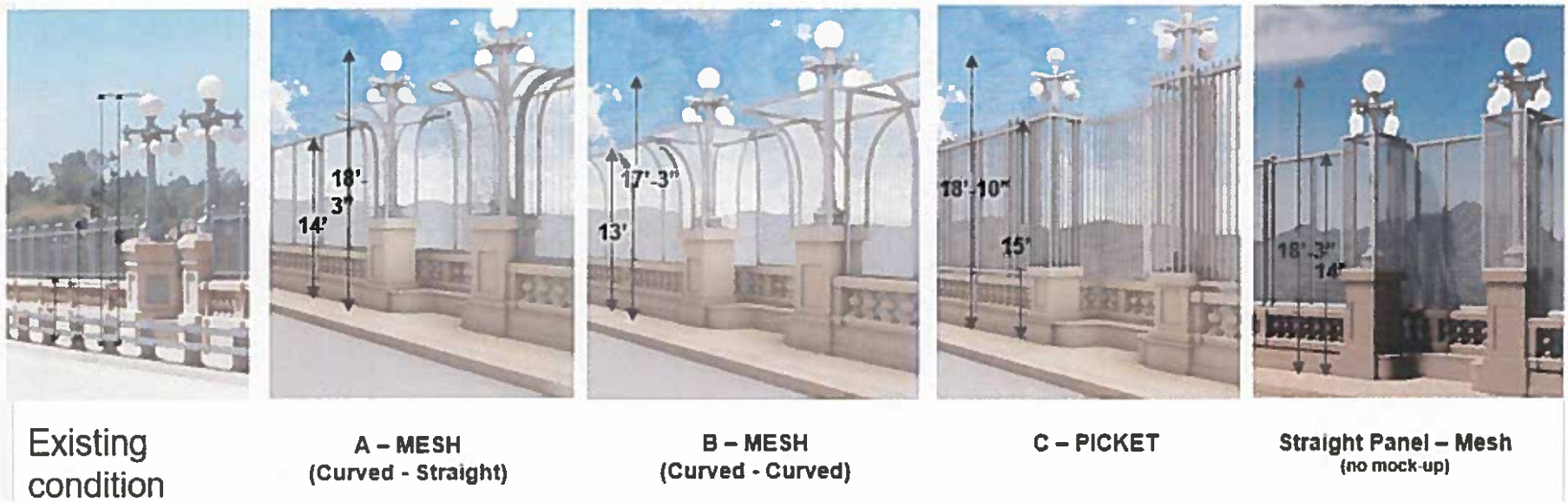
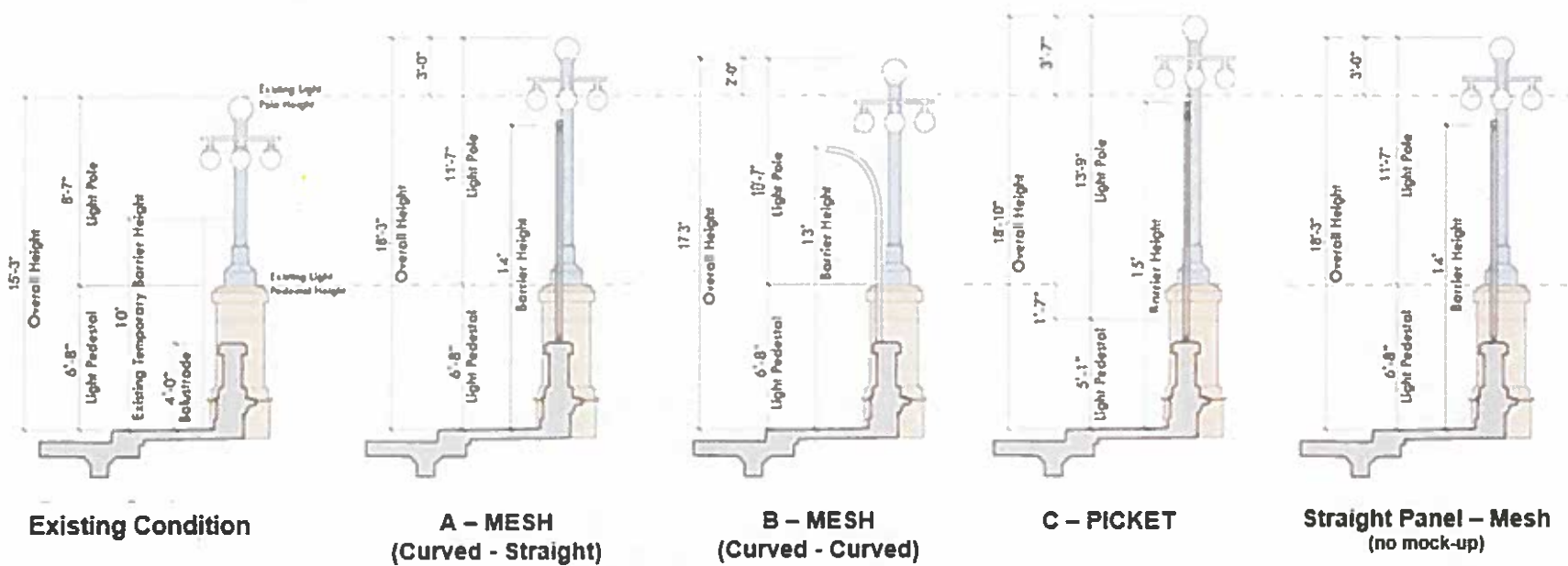
ALCOVE PERSPECTIVE



DONALD MACDONALD
ARCHITECTS LLP

Attachment D– Comparative Tables of Options

COMPARATIVE HEIGHTS OF THE OPTIONS



COMPARATIVE TABLE

Summary of options and their key attributes as it relates to the guiding principles for design and consideration.

	Option A (Mesh) Curved-Straight	Option B (Mesh) Curved-Curved	Option C (Picket)	Straight Panel: (Mesh) [no mock-up]
Renderings				
Effectiveness	• Top of straight barrier 10' to inhibit reach • Climb resistant mesh • Toeholds/handholds eliminated above balustrade and at lamp pilasters	• Curved barrier most effective to inhibit reach • Climb resistant mesh • Toeholds/handholds eliminated above balustrade and at lamp pilasters	• Distance between foot hold and toehold is 10' to inhibit reach • Pickets provide minimum climb resistance	• Top of straight barrier 10' to inhibit reach • Climb resistant mesh • Toeholds/handholds eliminated above balustrade and at lamp pilasters
Historic Preservation - (Character Defining Features)	• Pilasters remain at existing height • Light poles lengthened by 3 feet	• Pilasters remain at existing height • Light poles lengthened by 2 feet	• Pilasters lowered 1'-7" • Light poles lengthened 5'-2"	• Pilasters remain at existing height • Light poles lengthened by 3 feet
Aesthetics	• Transparency maintained with mesh fencing materials • Curves at lamp poles compliment the Beaux Arts • Vertical posts line up with bridge corbels	• Transparency maintained with mesh fencing material • Curves at lamp poles compliment the Beaux Arts • Vertical posts aligns with bridge corbels	• Transparency compromised with pickets stacking • Higher maintenance materials (requires painting)	• Transparency maintained with mesh fencing materials • Curves at lamp poles compliment the Beaux Arts • Vertical posts line up with bridge corbels

REFINED OPTIONS & ALTERNATIVE CONCEPT COMPARISONS



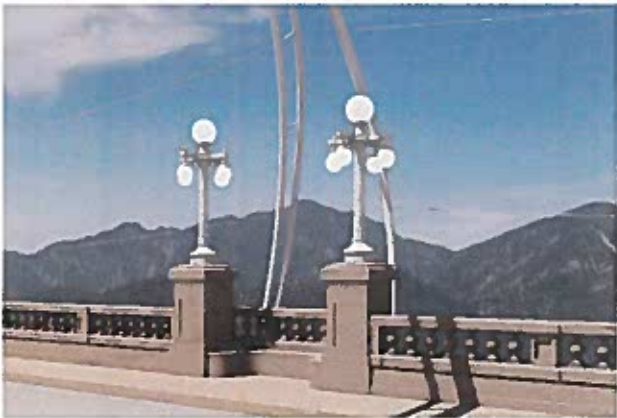
REFINED OPTION 01 - GLAZED ALCOVE & MESH PANELS

- REBUILT LIGHTPOST
- MESH INFILL
- BALUSTRADE CORBEL BLOCK POST
- GLAZING CONNECTED TO REINFORCED POSTS
- PANEL RAISED OFF OF EXISTING BALUSTRADE



REFINED OPTION 02 - GLAZED ALCOVE & TENSIONED RODS

- REBUILT LIGHTPOST
- TENSIONED ROD PANEL
- BALUSTRADE CORBEL BLOCK POST
- WRAPPED PILASTER GLAZING SUPPORTED ON POINT SUPPORTS
- GLAZED INFILL PANEL ON SHOE AND PINCH PLATES



CABLE NET ENCLOSURE SUPPORTED ON STEEL HALOS

- MULTI-PIPE STEEL 'HALO' BIFURCATED AT ALCOVE TO FRAME VIEWS OUT
- CABLE NET ENCLOSURE
- SUSPENSION CABLE
- EXISTING FENCE TO BE REMOVED
- LAMPS AND PILASTERS REMAIN AT EXISTING HEIGHT

COMPARATIVE HEIGHTS OF THE REFINED OPTIONS AND ALTERNATIVE CONCEPT

